

Window Functions – perform calculations across a set of rows relative to the current row.

Ranking Functions:

RANK()

DENSE RANK()

ROW\_NUMBER()

Aggregate Functions

SUM(), AVG, MIN(), MAX( ), COUNT()

Lead and Lag Functions

LEAD ( ) - accesses data from subsequent rows

LAG() – accesses data from preceding rows

Percentile Functions

PERCENT RANK() -relative rank within a group of values

CUME\_DIST() – cumulative distribution of a value within a group of values

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A simple Rank and Dense Rank example.

```
8  SELECT
9  company, widget_output,
10  RANK() OVER (ORDER BY widget_output DESC)
11  FROM widget_companies
12  ORDER BY widget_output DESC;
```

|   | company<br>text        | widget_output<br>integer | rank<br>bigint |
|---|------------------------|--------------------------|----------------|
| 1 | Miles Amalgamated      | 620000                   | 1              |
| 2 | Arthur Industries      | 244000                   | 2              |
| 3 | Fischer Worldwide      | 201000                   | 3              |
| 4 | Saito Widget Co.       | 201000                   | 3              |
| 5 | Dream Widget Inc.      | 196000                   | 5              |
| 6 | Ariadne Widget Masters | 143000                   | 6              |
| 7 | Mal Inc.               | 133000                   | 7              |
| 8 | Dom Widgets            | 125000                   | 8              |

This example creates ranks within groups of rows .  
(In this example ordering food categories for each store)

Each row here includes a store's product category and sales for that category.  
The SELECT statement creates a result showing how each store's sales ranks within each category. – The PARTITION BY OVER clause tells the in the first window (line 2) creates rankings one category at a time, using the store's unit sales in descending order.

The final ORDER BY clause includes the category column and the same RANK() function syntax.

```
1 SELECT category, store, unit_sales,  
2 rank() OVER(PARTITION BY category ORDER BY unit_sales DESC)  
3 FROM store_sales  
4 ORDER BY category, rank() OVER (PARTITION BY category  
5     ORDER BY unit_sales DESC)  
6 ;
```

Data Output Messages Notifications

SQL

Showing row

|   | category<br>[PK] text | store<br>[PK] text | unit_sales<br>bigint | rank<br>bigint |
|---|-----------------------|--------------------|----------------------|----------------|
| 1 | Beer                  | Wallace            | 988                  | 1              |
| 2 | Beer                  | Broders            | 641                  | 2              |
| 3 | Beer                  | Cramers            | 640                  | 3              |
| 4 | Cereal                | Broders            | 1104                 | 1              |
| 5 | Cereal                | Cramers            | 1003                 | 2              |
| 6 | Cereal                | Wallace            | 980                  | 3              |
| 7 | Ice Cream             | Broders            | 2517                 | 1              |
| 8 | Ice Cream             | Cramers            | 2112                 | 2              |
| 9 | Ice Cream             | Wallace            | 1863                 | 3              |

For reference here is the entire table without any filters:

```
13
14 SELECT *
15 FROM store_sales
16 ;
```

|   | store<br>[PK] text | category<br>[PK] text | unit_sales<br>bigint |
|---|--------------------|-----------------------|----------------------|
| 1 | Broders            | Cereal                | 1104                 |
| 2 | Wallace            | Ice Cream             | 1863                 |
| 3 | Broders            | Ice Cream             | 2517                 |
| 4 | Cramers            | Ice Cream             | 2112                 |
| 5 | Broders            | Beer                  | 641                  |
| 6 | Cramers            | Cereal                | 1003                 |
| 7 | Cramers            | Beer                  | 640                  |
| 8 | Wallace            | Cereal                | 980                  |
| 9 | Wallace            | Beer                  | 988                  |